Analysis I

Uebungsblatt 6

Problem:	Comment:	Points:
1		28 / 32

1

(a)

Series: $\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k}$

Solution. We apply the Leibniz criterion (alternating series test) to show convergence.

The series has the form $\sum_{k=0}^{\infty}(-1)^k a_k$ where $a_k = \frac{1}{2^k}$.

We need to verify three conditions:

1. Alternating signs: The factor $(-1)^k$ ensures alternating signs.

2. Terms decreasing in absolute value: We have $|a_k| = \frac{1}{2^k}$. For $k \geq 0$: $|a_{k+1}| = \frac{1}{2^{k+1}} = \frac{1}{2} \cdot \frac{1}{2^k} = \frac{1}{2}|a_k| < |a_k|$

Therefore $\{|a_k|\}$ is monotone decreasing.

3. Terms approach zero: $\lim_{k\to\infty} |a_k| = \lim_{k\to\infty} \frac{1}{2^k} = 0$

Since all three conditions of the Leibniz criterion are satisfied, the series $\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k}$ converges.

□

(b)

Series: $\sum_{k=1}^{\infty} \frac{1}{4k^2-1}$

Solution. We use the Direct Comparison Test to show convergence.

Step 1: Choose comparison series We know that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges (p-series with $p = 2 > 1$).

We will compare our series with $\sum_{k=1}^{\infty} \frac{1}{3k^2}$, which also converges since it equals $\frac{1}{3}\sum_{k=1}^{\infty} \frac{1}{k^2}$.

Step 2: Establish the inequality For $k \geq 1$, we need to show that $\frac{1}{4k^2-1} \leq \frac{1}{3k^2}$.

This is equivalent to showing $3k^2 \leq 4k^2 - 1$, or $1 \leq k^2$.

Since $k \geq 1$, we have $k^2 \geq 1$, so:

$$4k^2 - 1 \geq 4k^2 - k^2 = 3k^2$$

Taking reciprocals (and flipping the inequality):

$$\frac{1}{4k^2-1} \leq \frac{1}{3k^2}$$

Step 3: Apply the Direct Comparison Test Since:

- $\sum_{k=1}^{\infty} \frac{1}{3k^2}$ converges, and
- $0 < \frac{1}{4k^2-1} \leq \frac{1}{3k^2}$ for all $k \geq 1$

By the Direct Comparison Test (Theorem 3.5.4), the series $\sum_{k=1}^{\infty} \frac{1}{4k^2-1}$ converges. □

(c)

Series: $\sum_{k=1}^{\infty} \frac{k^4}{3^k}$

Solution. We use the Ratio Test to show convergence. This series has the form of a polynomial in the numerator and an exponential in the denominator, which suggests the Ratio Test.

Step 1: Apply the Ratio Test Let $a_k = \frac{k^4}{3^k}$. We compute:

$$\left|\frac{a_{k+1}}{a_k}\right| = \left|\frac{\frac{(k+1)^4}{3^{k+1}}}{\frac{k^4}{3^k}}\right| = \frac{(k+1)^4}{3^{k+1}} \cdot \frac{3^k}{k^4} = \frac{(k+1)^4}{k^4} \cdot \frac{3^k}{3^{k+1}} = \frac{(k+1)^4}{k^4} \cdot \frac{1}{3}$$

Step 2: Simplify the ratio

$$\left|\frac{a_{k+1}}{a_k}\right| = \frac{1}{3} \cdot \left(\frac{k+1}{k}\right)^4 = \frac{1}{3} \cdot \left(1 + \frac{1}{k}\right)^4$$

Step 3: Take the limit

$$\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty} \frac{1}{3} \cdot \left(1 + \frac{1}{k}\right)^4 = \frac{1}{3} \cdot (1+0)^4 = \frac{1}{3}$$

Step 4: Apply the Ratio Test criterion Since $\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \frac{1}{3} < 1$, the series $\sum_{k=1}^{\infty} \frac{k^4}{3^k}$ converges.

□

(d)

Series: $\sum_{k=1}^{\infty} \frac{k!}{k^k}$

Solution. We use the Ratio Test to show convergence. This series involves factorials, which makes the Ratio Test a natural choice.

Step 1: Set up the ratio Let $a_k = \frac{k!}{k^k}$. We compute:

$$\left|\frac{a_{k+1}}{a_k}\right| = \frac{\frac{(k+1)!}{(k+1)^{k+1}}}{\frac{k!}{k^k}} = \frac{(k+1)! \cdot k^k}{k! \cdot (k+1)^{k+1}}$$

Step 2: Simplify using factorial properties Since $(k+1)! = (k+1) \cdot k!$:

$$\frac{(k+1)! \cdot k^k}{k! \cdot (k+1)^{k+1}} = \frac{(k+1) \cdot k! \cdot k^k}{k! \cdot (k+1)^{k+1}} = \frac{(k+1) \cdot k^k}{(k+1)^{k+1}}$$

Step 3: Simplify the powers Since $(k+1)^{k+1} = (k+1) \cdot (k+1)^k$:

$$\frac{(k+1) \cdot k^k}{(k+1)^{k+1}} = \frac{(k+1) \cdot k^k}{(k+1) \cdot (k+1)^k} = \frac{k^k}{(k+1)^k} = \left(\frac{k}{k+1}\right)^k$$

Step 4: Rewrite for limit calculation

$$\left(\frac{k}{k+1}\right)^k = \left(\frac{1}{1+\frac{1}{k}}\right)^k = \frac{1}{\left(1+\frac{1}{k}\right)^k}$$

Step 5: Take the limit Using the fundamental limit $\lim_{k\to\infty}\left(1+\frac{1}{k}\right)^k = e$:

$$\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty}\frac{1}{\left(1+\frac{1}{k}\right)^k} = \frac{1}{e}$$

Step 6: Apply the Ratio Test criterion Since $\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| = \frac{1}{e} \approx 0.368 < 1$, the series $\sum_{k=1}^{\infty}\frac{k!}{k^k}$ converges.

□

2

(a)

Problem: Let $(a_n)_{n\in\mathbb{N}}$ be a sequence and $(a_{n_k})_{k\in\mathbb{N}}$ be a subsequence. Show by counterexample: If $\sum_{n\in\mathbb{N}} a_n$ converges, it does not generally follow that $\sum_{k\in\mathbb{N}} a_{n_k}$ converges.

Solution. We construct a counterexample using the alternating harmonic series.

Step 1: Define the sequence Consider the alternating harmonic series:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

This series converges by the Leibniz criterion (alternating series test):

- Terms alternate in sign: $(-1)^{n+1}$
- $\frac{1}{n} \to 0$ as $n \to \infty$
- $\frac{1}{n+1} < \frac{1}{n}$ (decreasing)

Step 2: Choose an appropriate subsequence Define the subsequence by taking only the positive terms:

$$n_k = 2k - 1 \text{ for } k = 1, 2, 3, \ldots$$

This gives us $n_1 = 1, n_2 = 3, n_3 = 5, n_4 = 7, \ldots$ (all odd indices).

Step 3: Examine the subsequence series The corresponding subsequence series is:

$$\sum_{k=1}^{\infty} a_{n_k} = \sum_{k=1}^{\infty} a_{2k-1} = \sum_{k=1}^{\infty}\frac{(-1)^{2k}}{2k-1} = \sum_{k=1}^{\infty}\frac{1}{2k-1}$$

This becomes:

$$\frac{1}{1}+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+\frac{1}{9}+\cdots$$

Step 4: Show the subsequence series diverges This is the series of reciprocals of odd numbers. We can show it diverges by comparison with the harmonic series:

Since $\frac{1}{2k-1} > \frac{1}{2k}$ for all $k \geq 1$, we have:

$$\sum_{k=1}^{\infty}\frac{1}{2k-1} > \sum_{k=1}^{\infty}\frac{1}{2k} = \frac{1}{2}\sum_{k=1}^{\infty}\frac{1}{k}$$

Since $\sum_{k=1}^{\infty}\frac{1}{k}$ diverges (harmonic series), the series $\sum_{k=1}^{\infty}\frac{1}{2k-1}$ also diverges.

Conclusion: We have shown that:

- $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$ converges
- $\sum_{k=1}^{\infty} a_{n_k} = \sum_{k=1}^{\infty}\frac{1}{2k-1}$ diverges

Therefore, convergence of the original series does not imply convergence of every subsequence series. □

(b)

Problem: Prove: If $\sum_{n\in\mathbb{N}} a_n$ converges absolutely, then $\sum_{k\in\mathbb{N}} a_{n_k}$ also converges absolutely.

Solution. We prove this using the Monotone Bounded Theorem (Satz 3.2.2).

Given: $\sum_{n=1}^{\infty} a_n$ converges absolutely, so $\sum_{n=1}^{\infty}|a_n|$ converges.

To prove: $\sum_{k=1}^{\infty} a_{n_k}$ converges absolutely, i.e., $\sum_{k=1}^{\infty}|a_{n_k}|$ converges.

Step 1: Define partial sums Let $S_N = \sum_{n=1}^{N}|a_n|$ and $T_K = \sum_{k=1}^{K}|a_{n_k}|$

Since $\sum_{n=1}^{\infty}|a_n|$ converges, we have $\lim_{N\to\infty} S_N = S < \infty$.

Step 2: Show (T_K) is bounded above Since (n_k) is a strictly increasing subsequence with $n_1 < n_2 < n_3 < \cdots$, each term $|a_{n_k}|$ appears in the partial sum S_{n_K} of the absolutely convergent series.

Therefore:

$$T_K = |a_{n_1}| + |a_{n_2}| + \cdots + |a_{n_K}| \leq S_{n_K} \leq S < \infty$$

So the sequence (T_K) is bounded above by S.

Step 3: Show (T_K) is monotone increasing Since all terms $|a_{n_k}| \geq 0$:

$$T_{K+1} = T_K + |a_{n_{K+1}}| \geq T_K$$

Therefore (T_K) is monotone increasing.

?

Step 4: Apply the Monotone Bounded Theorem Since (T_K) is monotone increasing and bounded above, it converges:

-1P

$$\lim_{K\to\infty} T_K = \lim_{K\to\infty} \sum_{k=1}^{K} |a_{n_k}| \text{ exists and is finite}$$

This means $\sum_{k=1}^{\infty} |a_{n_k}|$ converges.

Step 5: Conclude absolute convergence Since $\sum_{k=1}^{\infty} |a_{n_k}|$ converges, the series $\sum_{k=1}^{\infty} a_{n_k}$ converges absolutely.

□

(c)

Task: Using part (b), compute the sum of the series in Problem 1(a):

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k}.$$

Solution. By part (b), if a series is absolutely convergent, then any subsequence series is also absolutely convergent and we may split the series into disjoint subsequences and sum them. Since

$$\sum_{k=0}^{\infty} \left| \frac{(-1)^k}{2^k} \right| = \sum_{k=0}^{\infty} \frac{1}{2^k}$$

converges, the original series is absolutely convergent. Split it into the even and the odd subsequences:

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k} = \sum_{m=0}^{\infty} \frac{1}{2^{2m}} + \sum_{m=0}^{\infty} \frac{(-1)}{2^{2m+1}}.$$

Both sums are geometric:

$$\sum_{m=0}^{\infty} \frac{1}{2^{2m}} = \sum_{m=0}^{\infty} \left(\frac{1}{4}\right)^m = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}, \qquad \sum_{m=0}^{\infty} \frac{(-1)}{2^{2m+1}} = -\frac{1}{2} \sum_{m=0}^{\infty} \left(\frac{1}{4}\right)^m = -\frac{1}{2} \cdot \frac{1}{1-\frac{1}{4}} = -\frac{2}{3}.$$

Adding the two convergent subsequence sums yields

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k} = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}.$$

□

3

Problem: Let $(x_n)_{n\in\mathbb{N}}$ be a real sequence with $L := \limsup_{n\to\infty} x_n > -\infty$. Show that for every $\varepsilon > 0$ there are at most finitely many indices k such that

$$x_k \geq L + \varepsilon.$$

Solution. **Solution via tail suprema.** Recall that $L = \lim_{n\to\infty} s_n$ where $s_n := \sup_{k>n} x_k$ and that (s_n) is decreasing. Fix $\varepsilon > 0$ and set $\delta := \varepsilon/2$. Since $s_n \to L$, there exists N such that for all $n \geq N$ we have

$$s_n < L + \delta = L + \varepsilon/2.$$

Assume by contradiction that there are infinitely many k with $x_k \geq L + \varepsilon$. Then we can choose some $k \geq N$ with this property. But for any $k \geq N$ we also have

$$x_k \leq s_k \leq s_N < L + \varepsilon/2,$$

which contradicts $x_k \geq L + \varepsilon$. Hence only finitely many indices can satisfy $x_k \geq L + \varepsilon$.

□

4

Problem: Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of nonnegative real numbers with $\sum_{n=1}^{\infty} a_n < \infty$. Show that the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n}$$

converges.

where from? Proof -3P

Solution. Set $x_n := \sqrt{a_n}$ and $y_n := \frac{1}{n}$. By the Cauchy–Schwarz inequality for series,

$$\sum_{n=1}^{\infty} x_n y_n \leq \Big(\sum_{n=1}^{\infty} x_n^2\Big)^{1/2} \Big(\sum_{n=1}^{\infty} y_n^2\Big)^{1/2}.$$

Since $x_n^2 = a_n$ and $y_n^2 = 1/n^2$, we obtain

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n} \leq \Big(\sum_{n=1}^{\infty} a_n\Big)^{1/2} \Big(\sum_{n=1}^{\infty} \frac{1}{n^2}\Big)^{1/2} < \infty,$$

because $\sum a_n$ converges by assumption and $\sum 1/n^2$ converges (p-series with $p = 2$). Hence $\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n}$ converges absolutely.

□

Did you solve it yourself?

top

© Universität Zürich | Sep 24, 2025